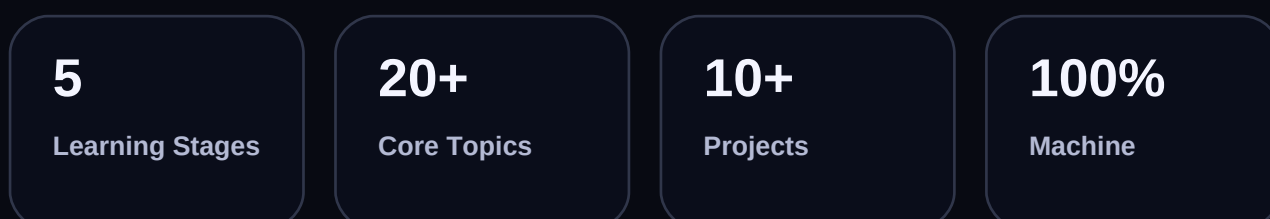
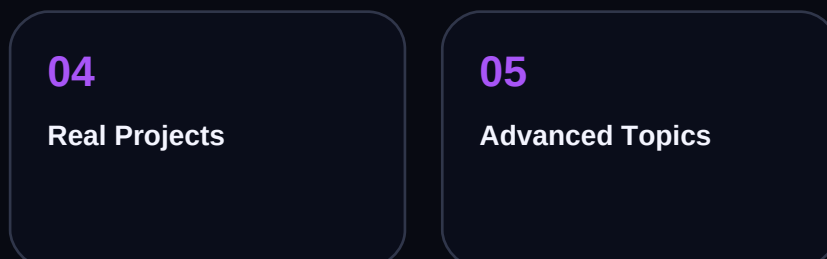
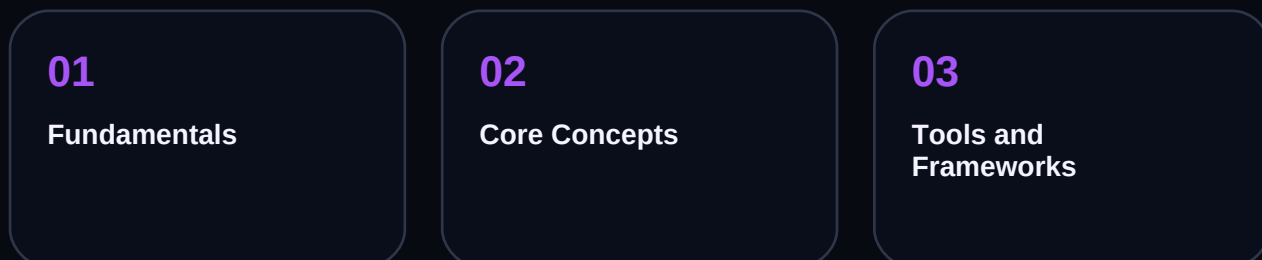
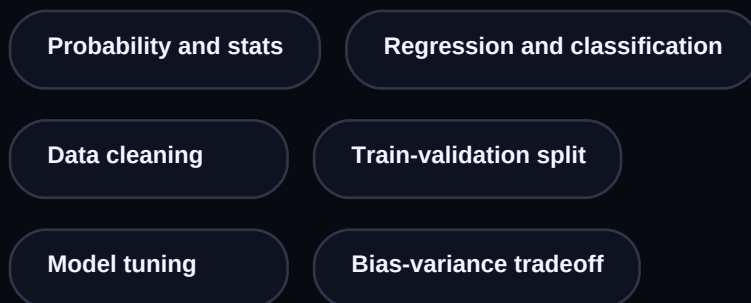


MACHINE LEARNING

Complete Learning Roadmap

Move from statistical learning basics to scalable model development and serving.





Learning Path Overview

Five progressive stages from fundamentals to production mastery

01

Fundamentals

Probability and stats •
Regression and classification •
Data cleaning • Train-validation
split

02

Core Concepts

Model tuning • Bias-variance
tradeoff • Feature selection •
Cross validation

03

Tools and Frameworks

scikit-learn • MLflow

04

Real Projects

Churn predictor • Fraud
detection baseline •
Recommendation prototype •
Model report card

05

Advanced Topics

Online inference optimization •
A/B model rollouts • Concept
drift controls • MLOps
governance

How to Use This Roadmap

Build and ship first

Complete a small working project in every stage before advancing.

Use official sources

Start with the docs and use secondary resources to reinforce implementation patterns.

Iterate on projects

Refactor earlier work as you learn better architecture, testing, and performance habits.

Stages 01 & 02 - Deep Dive

Fundamentals, core concepts, and architectural foundations

01

Stage 1 - Fundamentals

Estimated: 3-4 weeks | Prerequisites: Basic foundations

Topics

- Probability and stats
- Regression and classification
- Data cleaning
- Train-validation split

Tools & Frameworks

- Probability and stats
- Regression and classification
- Data cleaning
- Train-validation split

Recommended Projects

- Machine Learning workshop using Probability and stats
- Machine Learning portfolio project with Regression and classification

Stage 1 Milestone Complete Machine Learning workshop using Probability and stats and document the decisions, trade-offs, and validation checks before moving to Stage 2.

02

Stage 2 - Core Concepts

Estimated: 4-5 weeks | Prerequisites: Stage 1 complete

Topics

- Model tuning
- Bias-variance tradeoff
- Feature selection
- Cross validation

Tools & Frameworks

- Model tuning
- Bias-variance tradeoff
- Feature selection
- Cross validation

Recommended Projects

- Machine Learning workshop using Model tuning
- Machine Learning portfolio project with Bias-variance tradeoff

Stage 2 Milestone Complete Machine Learning workshop using Model tuning and document the decisions, trade-offs, and validation checks before moving to Stage 3.



Stages 03 & 04 - Deep Dive

Modern tooling, delivery frameworks, and portfolio projects

03

Stage 3 - Tools and Frameworks

Estimated: 4-6 weeks | Prerequisites: Stage 2 complete

Topics

- scikit-learn
- MLflow
- Pipeline automation
- Model registry

Tools & Frameworks

- scikit-learn
- MLflow

Recommended Projects

- Machine Learning workshop using scikit-learn
- Machine Learning portfolio project with MLflow

Stage 3 Milestone Complete Machine Learning workshop using scikit-learn and document the decisions, trade-offs, and validation checks before moving to Stage 4.

04

Stage 4 - Real Projects

Estimated: 6-8 weeks | Prerequisites: Stage 3 complete

Topics

- Churn predictor
- Fraud detection baseline
- Recommendation prototype
- Model report card

Tools & Frameworks

- Churn predictor
- Fraud detection baseline
- Recommendation prototype
- Model report card

Recommended Projects

- Machine Learning workshop using Churn predictor
- Machine Learning portfolio project with Fraud detection baseline

Stage 4 Milestone Complete Machine Learning workshop using Churn predictor and document the decisions, trade-offs, and validation checks before moving to Stage 5.

Stage 05 - Advanced Topics & Delivery

Performance, architecture, release automation, and production governance

05

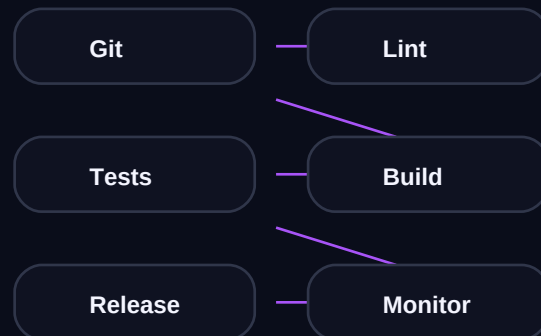
Stage 5 - Advanced Topics

Estimated: Ongoing | Industry-grade operating habits

Advanced Focus

- Online inference optimization
- A/B model rollouts
- Concept drift controls
- MLOps governance

CI/CD Pipeline



Stage 5 Milestone

A production-grade machine learning workflow with measurable performance, quality, and release confidence.

Performance Budget

Set and track concrete engineering thresholds: startup latency, error budget, throughput, cost efficiency, and deployment confidence.

Resources & Practice Questions

Verified links, self-assessment, and community follow-up

FREE

Machine Learning docs and community guides

Link: <https://scikit-learn.org/stable/>

FREE

Build one portfolio project per stage before moving ahead

Link: <https://roadmap.sh/ai-data-scientist>

PAID

Structured specialization track focused on Machine Learning

Link: <https://roadmap.sh/ai-data-scientist>

FREE

Machine Learning roadmap reference

Link: <https://roadmap.sh/ai-data-scientist>

FREE

Official documentation

Link: <https://scikit-learn.org/stable/>

Self-Assessment Practice Questions

- How would you evaluate readiness for Stage 3 in Machine Learning?
- Which project best proves Stage 4 competency in Machine Learning?
- What trade-offs appear in Stage 5 decisions for Machine Learning?